

depletion represents nearly 15% of the mean energy depletion among countries in the sample. Interestingly, the *ATT* on energy depletion is statistically significant at 10% level under *Spec1*, but not under *Spec2*, highlighting that propensity score misspecifications can also lead to invalid inference. Finally, we note that although our tests do not detect propensity score misspecification for the rural and urban water access, the results suggest that GATT/WTO membership is not statistically significant using either propensity score specification, perhaps because of the relatively high standard errors due to the limited sample size.

Overall, we find that propensity score misspecifications can affect both the economic and statistical conclusions about the effect of GATT/WTO membership on environmental quality. When the propensity score is correctly specified, our results suggest that GATT/WTO membership is associated with improved environmental performance in terms of CO_2 and energy depletion but not in terms of rural and urban water access. We also find that GATT/WTO membership is associated with higher deforestation, though the statistical evidence is relatively weaker. Furthermore, our results uncover interesting heterogeneity, as the aforementioned results are statistically significant for the overall population (*ATE*) but not for the treated-subpopulation (*ATT*).

6 Conclusion

In this article, we have shown that, when propensity scores are correctly specified, a particular restriction between the propensity score CDFs of treated and control groups must hold. Based on such restriction, we propose new nonparametric projection-based tests for the correct specification of the propensity score. In contrast to other proposals, our tests are not severely affected by the “curse of dimensionality”, are not sensitive to the different estimation methods used to estimate the propensity score under the null, do not rely on the potentially ad hoc choice of bandwidths, and enjoy some optimal power properties against particular classes of alternative hypotheses. We have derived the asymptotic properties of the proposed tests, and have proved that they are able to detect local alternatives converging to the null at the parametric rate, and that critical values can be easily computed via a simple multiplier bootstrap procedure. Our Monte Carlo simulation study illustrates that, for a large class of alternatives, our projection-based tests perform better in finite samples than existing tests, though there are some classes of alternatives in which our tests have trivial power. All these finite sample findings are in line